

Factorising a polynomial

Taking out the common factor, and grouping when there is no single one.

LESSON 77–78

2 × 40 MIN

MATEMATIKA 7, §36

S8 2.4

LESSON CLOCK

10'

25'

25'

25'

5'

The idea	10 min
The common factor	25 min
Grouping	25 min
What it is for	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find the highest common factor of the terms of a polynomial.
- Take that factor outside a bracket, and check by expanding.
- Factorise a four-term polynomial by grouping.
- Use a factorised form to evaluate an expression quickly.

TEXTBOOK

National	pp. 218–224
Cambridge	Stage 8, pp. 25–30
Workbook	Exercise 2.4

01

Explanation

The idea

Expanding turns a product into a sum. **Factorising** is the same road travelled backwards: it turns a sum into a product.

$$3x(2x + 3) = 6x^2 + 9x \rightarrow 6x^2 + 9x = 3x(2x + 3)$$

DIRECTION	START	FINISH
expanding	$3x(2x + 3)$	$6x^2 + 9x$
factorising	$6x^2 + 9x$	$3x(2x + 3)$

THE CHECK IS FREE

Every factorisation can be checked by expanding it again. If the expansion does not return the original polynomial exactly, the factorisation is wrong — there is nothing to argue about.

The common factor

Take the highest common factor of **all** the terms outside the bracket. For the numbers that is their HCF; for each letter it is the lowest power that appears in every term.

POLYNOMIAL	HCF	FACTORISED
$6x^2 + 9x$	$3x$	$3x(2x + 3)$
$8a^3 - 4a^2$	$4a^2$	$4a^2(2a - 1)$
$10x^3y - 15x^2y^2$	$5x^2y$	$5x^2y(2x - 3y)$
$12a^2b + 8ab^2 - 4ab$	$4ab$	$4ab(3a + 2b - 1)$

A TERM THAT IS ENTIRELY THE FACTOR LEAVES A 1

In $12a^2b + 8ab^2 - 4ab$ the last term is $-4ab$, which is the factor itself; inside the bracket it becomes -1 , not nothing. Losing that 1 is the commonest error of the whole topic, and expanding back catches it at once.

Grouping

When the four terms have no common factor between all of them, split them into two pairs that do, and a common bracket appears.

$$ax + ay + bx + by = a(x + y) + b(x + y) = (x + y)(a + b)$$

STEP	WORKING
the polynomial	$2x^2 + 6x + ax + 3a$
pair them	$(2x^2 + 6x) + (ax + 3a)$
factorise each pair	$2x(x + 3) + a(x + 3)$
the bracket is common	$(x + 3)(2x + a)$

A MINUS IN FRONT OF THE SECOND PAIR CHANGES BOTH SIGNS

$x^2 - 3x - 2x + 6$ pairs as $x(x - 3) - 2(x - 3)$, giving $(x - 3)(x - 2)$. Writing $-2(x + 3)$ there is a sign slip, and the two brackets then fail to match.

What it is for

A product is easier to evaluate, easier to cancel and — next year — the only way to solve an equation.

USE	EXAMPLE	WHY THE PRODUCT HELPS
evaluating	$37 \cdot 24 + 37 \cdot 76$	$37(24 + 76) = 3700$
cancelling	$\frac{6x^2 + 9x}{3x}$	$\frac{3x(2x + 3)}{3x} = 2x + 3$
divisibility	$n^2 + n$	$n(n + 1)$

WHY $n^2 + n$ IS ALWAYS EVEN

Factorised it is $n(n + 1)$ — the product of two consecutive whole numbers, one of which must be even. In the unfactorised form the fact is invisible; that is the whole value of a product.

02 Worked examples

EXAMPLE 1 Factorise $12a^2b + 8ab^2 - 4ab$.

- ① HCF of 12, 8, 4 is 4.
- ② Every term has a and b , so the HCF is $4ab$.
- ③ $= 4ab(3a + 2b - 1)$
The last term leaves -1 .
- ④ Check: $4ab(3a + 2b - 1) = 12a^2b + 8ab^2 - 4ab$ ✓

ANSWER

$$4ab(3a + 2b - 1)$$

EXAMPLE 2 Factorise $2x^2 + 6x + ax + 3a$ by grouping.

$$(1) \quad (2x^2 + 6x) + (ax + 3a)$$

$$(2) \quad 2x(x + 3) + a(x + 3)$$

The same bracket in both.

$$(3) \quad = (x + 3)(2x + a)$$

$$(4) \quad \text{Check: } (x + 3)(2x + a) = 2x^2 + ax + 6x + 3a \quad \checkmark$$

ANSWER

$$(x + 3)(2x + a)$$

EXAMPLE 3 Evaluate $37 \cdot 24 + 37 \cdot 76$ without a calculator.

$$(1) \quad \text{Both terms contain 37.}$$

$$(2) \quad = 37(24 + 76)$$

$$(3) \quad = 37 \cdot 100$$

$$(4) \quad = 3700$$

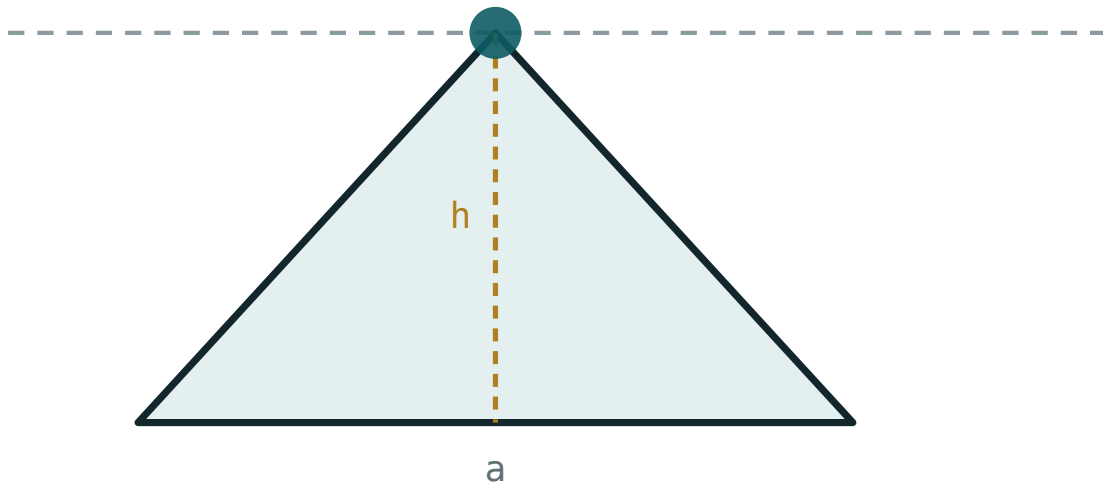
ANSWER

$$3700$$

03 Interactive model

Give the class the factorised answers first and ask them to expand; when they then meet the sums, the factor is already familiar and the missing 1 stops appearing.

A common factor as one side of a rectangle

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Factor	Ko'paytuvchi	Множитель
Highest common factor	Eng katta umumiy bo'luvchi	Наибольший общий делитель
To take outside the bracket	Qavsdan tashqariga chiqarish	Вынести за скобку
Grouping	Guruhlash	Группировка
Product	Ko'paytma	Произведение
Sum	Yig'indi	Сумма
Check by expanding	Qavsni ochib tekshirish	Проверка раскрытием

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Factorising turns a sum into:

a difference

a product

a quotient

a power

$6x^2 + 9x$ factorises to:

$3x(2x + 3)$

$3x(2x + 9)$

$x(6x + 9)$

$3(2x^2 + 3x)$

In $12a^2b + 8ab^2 - 4ab$ the last term leaves:

0

-1

-4ab

1

Grouping is used when:

there are two terms

no factor is common to all

the terms are like

there is a power

$ax + ay + bx + by$ equals:

$(x + y)(a + b)$

$(a + x)(b + y)$

$ab(x + y)$

$(a + b)^2$

A factorisation is checked by:

dividing

expanding

substituting 0

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $4x + 8$

ANSWER $4(x + 2)$

2. $6x^2 + 9x$

ANSWER $3x(2x + 3)$

3. $8a^3 - 4a^2$

ANSWER $4a^2(2a - 1)$

4. $5x - 15$

ANSWER $5(x - 3)$

5. $x^2 + x$

ANSWER $x(x + 1)$

6. $9y^3 + 3y$

ANSWER $3y(3y^2 + 1)$

7. $10ab - 5a$

ANSWER $5a(2b - 1)$

Medium · the standard question

7 PROBLEMS

1. $10x^3y - 15x^2y^2$

ANSWER $5x^2y(2x - 3y)$

2. $12a^2b + 8ab^2 - 4ab$

ANSWER $4ab(3a + 2b - 1)$

3. $ax + ay + bx + by$

ANSWER $(x + y)(a + b)$

4. $2x^2 + 6x + ax + 3a$

ANSWER $(x + 3)(2x + a)$

5. $x^2 - 3x - 2x + 6$

ANSWER $(x - 3)(x - 2)$

6. $37 \cdot 24 + 37 \cdot 76$

ANSWER 3700

7. $14m^2n - 21mn^2$

ANSWER $7mn(2m - 3n)$

Hard · stretch and reason

7 PROBLEMS

1. $15x^4y^2 - 10x^3y^3 + 5x^2y$

ANSWER $5x^2y(3x^2y - 2xy^2 + 1)$

2. $3a(x - 2) + 5(x - 2)$

ANSWER $(x - 2)(3a + 5)$

3. $x(a - b) - y(b - a)$

ANSWER $(a - b)(x + y)$

4. Show that $n^2 + n$ is always even

ANSWER $n(n + 1)$ — one of them is even

5. $56 \cdot 19 + 56 \cdot 81$

ANSWER 5600

6. Simplify $\frac{4a^2b - 6ab^2}{2ab}$ by factorising first

ANSWER $2a - 3b$

7. Factorise $x^3 + x^2 + x + 1$

ANSWER $(x + 1)(x^2 + 1)$

07 Homework**Homework — 5 tasks**

Expand every answer back; a factorisation that does not return the original is not an answer.

- Factorise $12x^2 - 8x$ and $9a^3b - 6a^2b^2$.
- Factorise $6m^2n + 9mn^2 - 3mn$, and say what the last term leaves.
- Factorise $3x^2 + 12x + ax + 4a$ by grouping.
- Evaluate $48 \cdot 37 + 48 \cdot 63$ by factorising.

5. Explain, using a factorisation, why $n^2 - n$ is always even.